Information Technology

Fall 2026-2027

Embedded Systems
LAB 2

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الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

Grading Scheme

Quiz 1	Quiz 2	Assignment	Section Attendance	Project	Mid-term Exam	Final Exam
5 points	5 points	5 points	5 points	10 points	20 points	50 points

Total: **100 Points**

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Objective

- This is the first program to run on the LaunchPad
- We will run this program without modification.

The logic of the program is:

If the left switch SW1 is not pressed

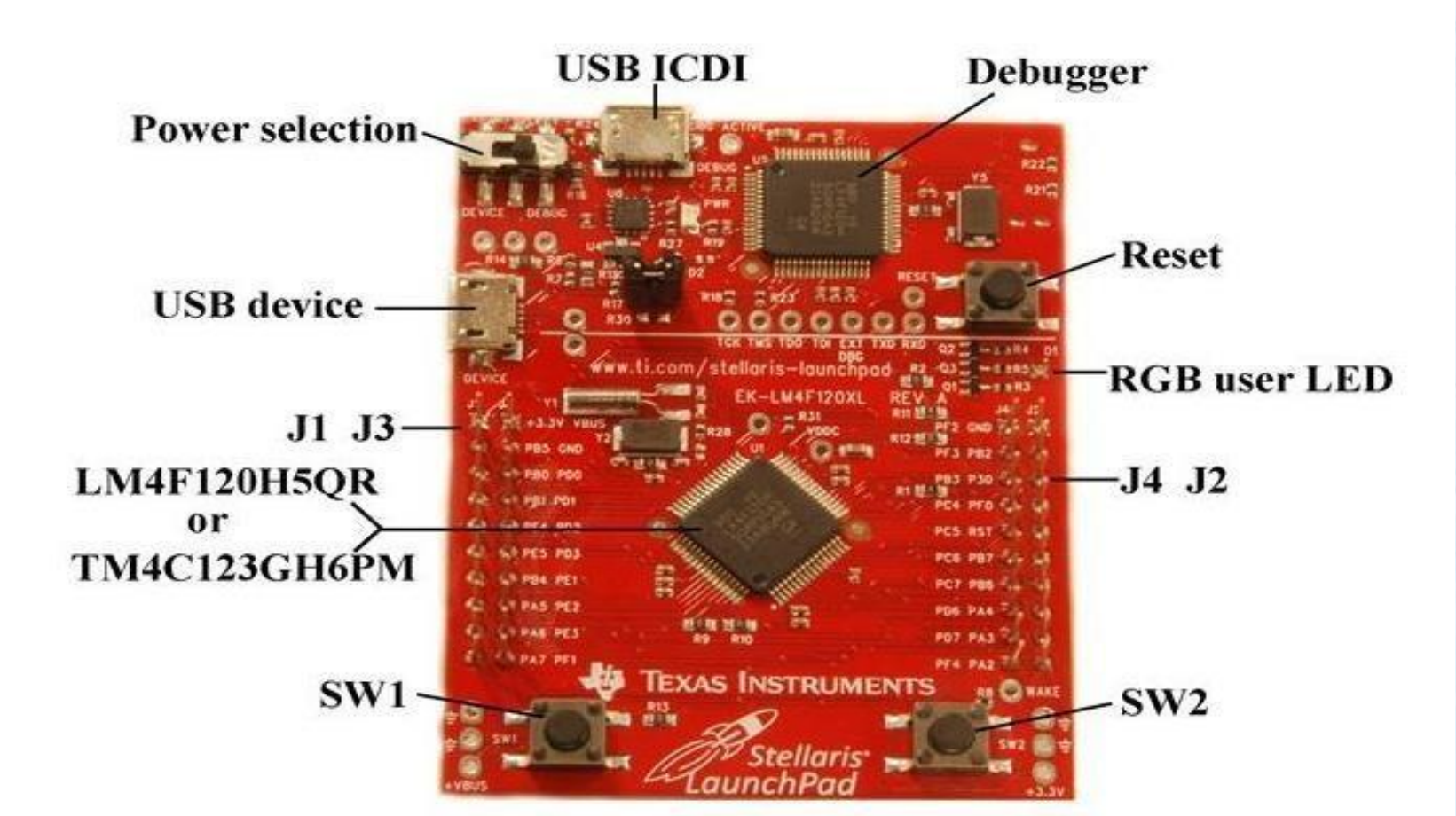
the LED toggles **blue**-**red**

If the left switch SW1 is Pressed

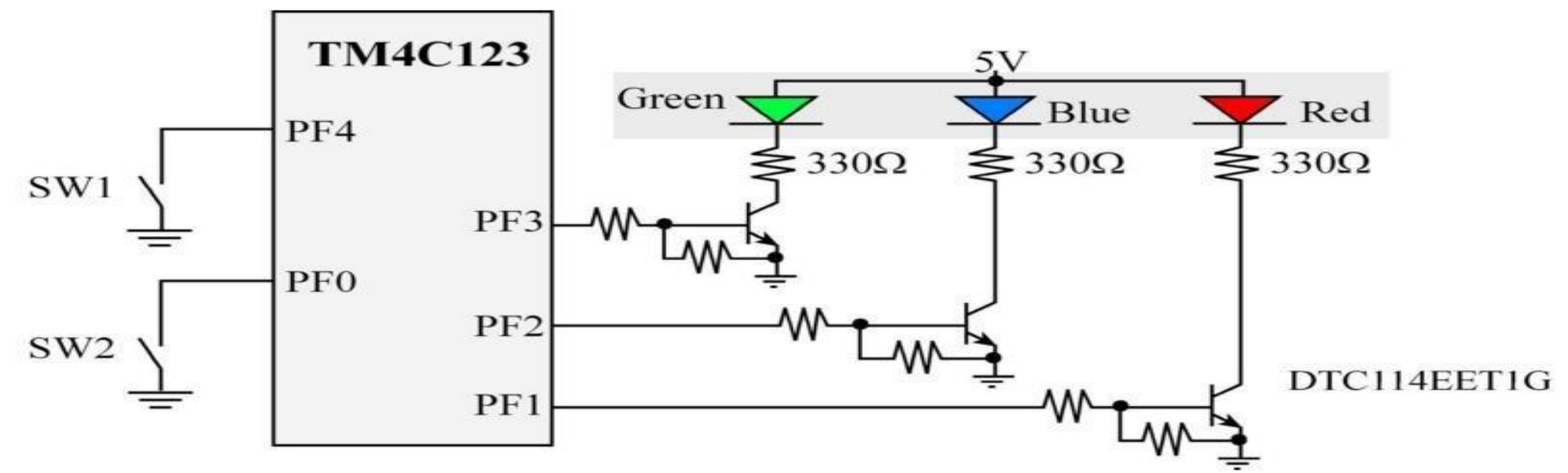
the LED toggles **blue**-**Green**

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Microcontroller ARM



Microcontroller input/output port



Notice if SW1 pressed (closed) the PF4 will connect to Earth → PF4 = 0

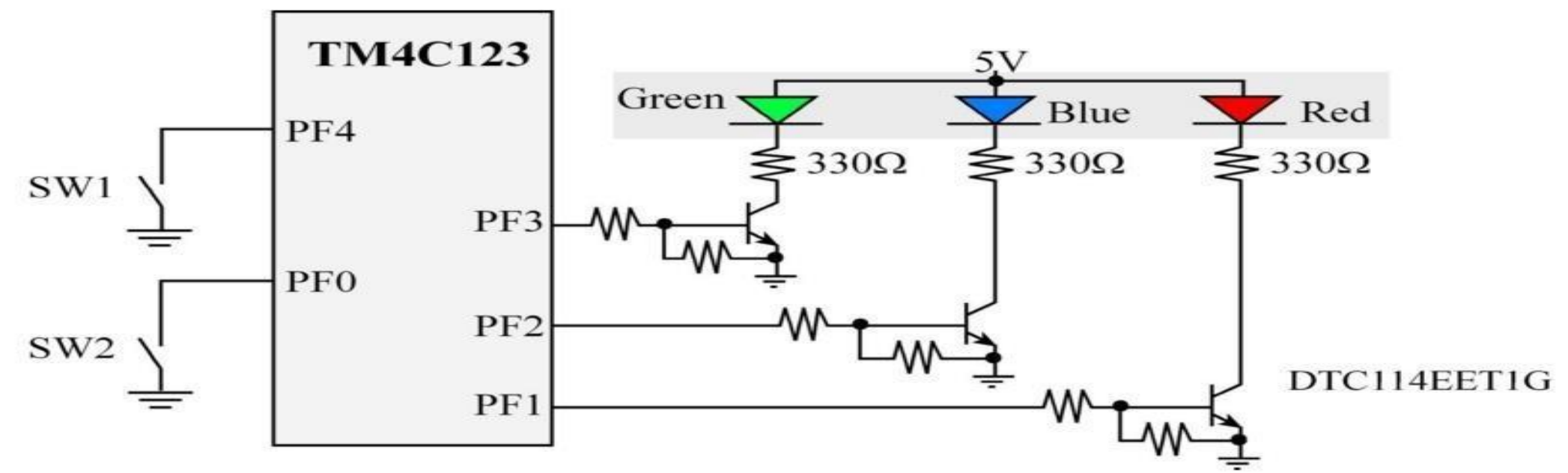
Notice if SW1 not pressed (Open) the PF4 will be floating → PF4 = 1

(Negative Logic)

Same with SW2 and PF0

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Microcontroller input/output port



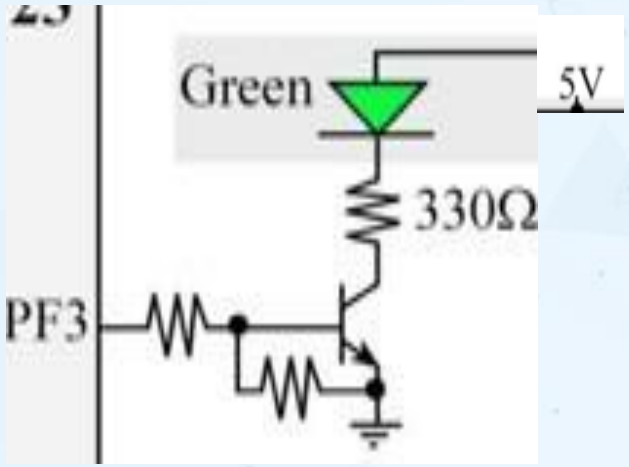
RED LED is connected to PF1

BLUE LED is connected to PF2

Green LED is connected to PF3

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Microcontroller input/output port



The circuit of Green LED and PF3

The transistor has input at collector 5 v

IF PF3 = 1 (5 v) .. The switch(transistor) will be closed

So LED is Lighted. (Positive Logic)

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Values of port

			SW1	GREEN LED	BLUE LED	RED LED	SW2
PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0

PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0	Hexadecimal
0	0	0	0	0	0	0	0	X00
0	0	0	0	0	0	0	1	X 01
0	0	0	0	0	0	1	0	X 02
0	0	0	0	0	0	1	1	X 03
0	0	0	0	0	1	0	0	X 04
0	0	0	0	0	1	0	1	X 05
0	0	0	0	0	1	1	0	X 06
0	0	0	0	0	1	1	1	X 07

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Values of port

			SW1	GREEN LED	BLUE LED	RED LED	SW2
PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0

PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0	Hexadecimal
0	0	0	0	1	0	0	0	X08
0	0	0	0	1	0	0	1	X 09
0	0	0	0	1	0	1	0	X 0A
0	0	0	0	1	0	1	1	X 0B
0	0	0	0	1	1	0	0	X 0C
0	0	0	0	1	1	0	1	X 0D
0	0	0	0	1	1	1	0	X 0E
0	0	0	0	1	1	1	1	X 0F

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Values of port

			SW1	GREEN LED	BLUE LED	RED LED	SW2
PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0

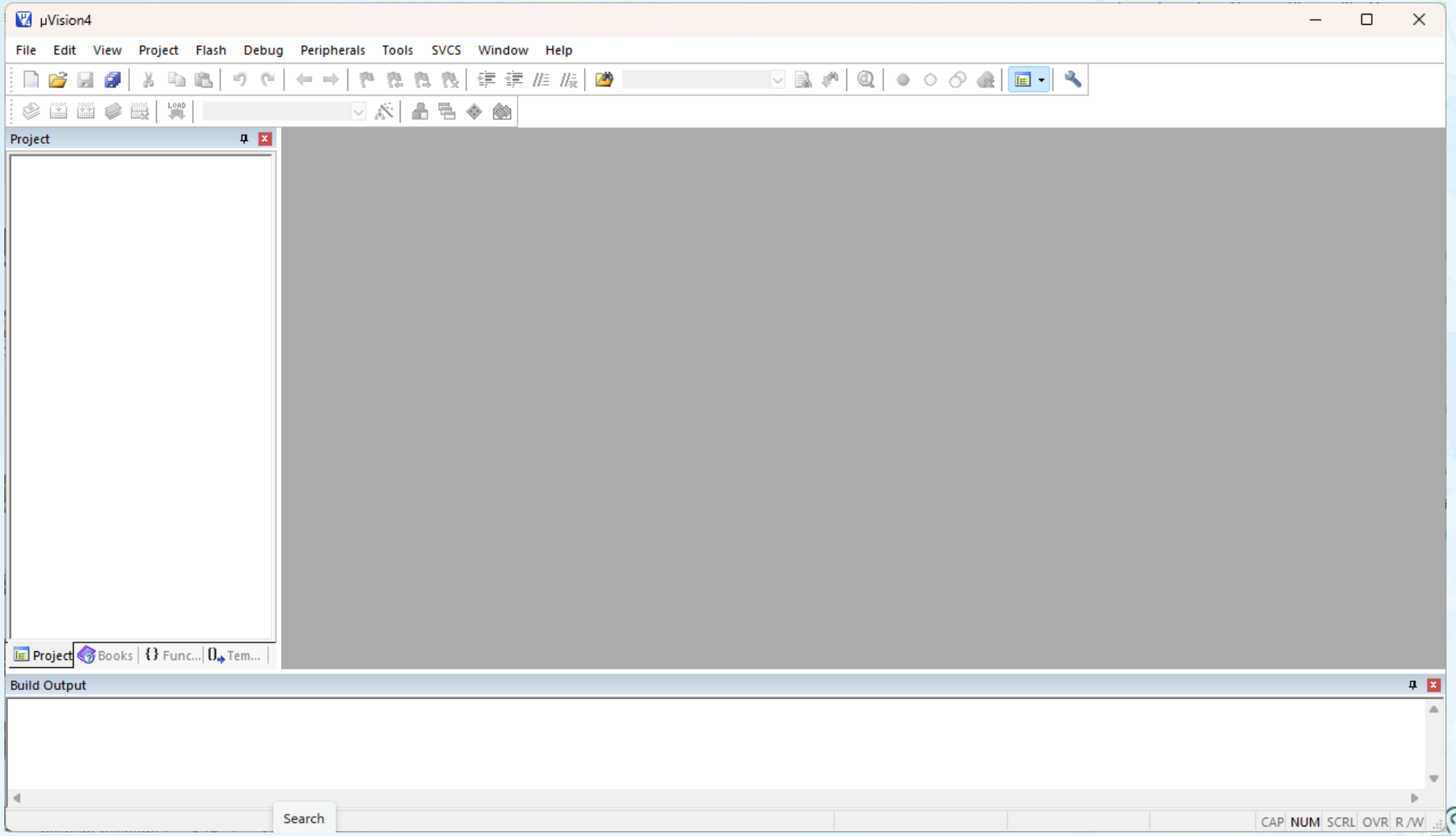
PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0	Hexadecimal
1	1	1	1	1	0	0	0	XF8
1	1	1	1	1	0	0	1	X F9
1	1	1	1	1	0	1	0	X FA
1	1	1	1	1	0	1	1	X FB
1	1	1	1	1	1	0	0	X FC
1	1	1	1	1	1	0	1	X FD
1	1	1	1	1	1	1	0	X FE
1	1	1	1	1	1	1	1	X FF

Values of port

			SW1	GREEN LED	BLUE LED	RED LED	SW2
PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0

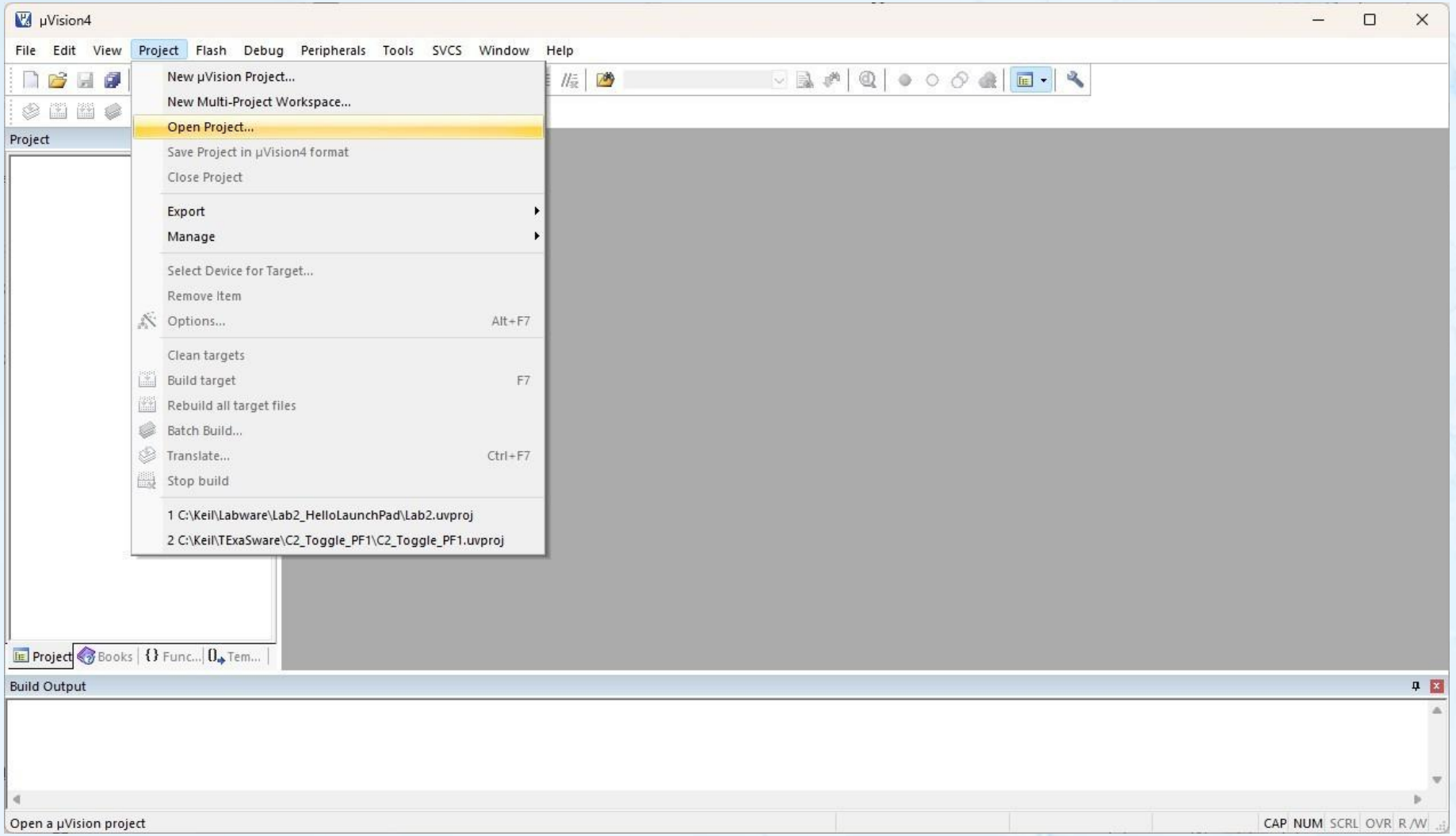
			PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0
// red	R	0x02	0	0	0	0	0	0	1	0
// blue	B	0x04	0	0	0	0	0	1	0	0
// green	G	0x08	0	0	0	0	1	0	0	0

First Step: Open

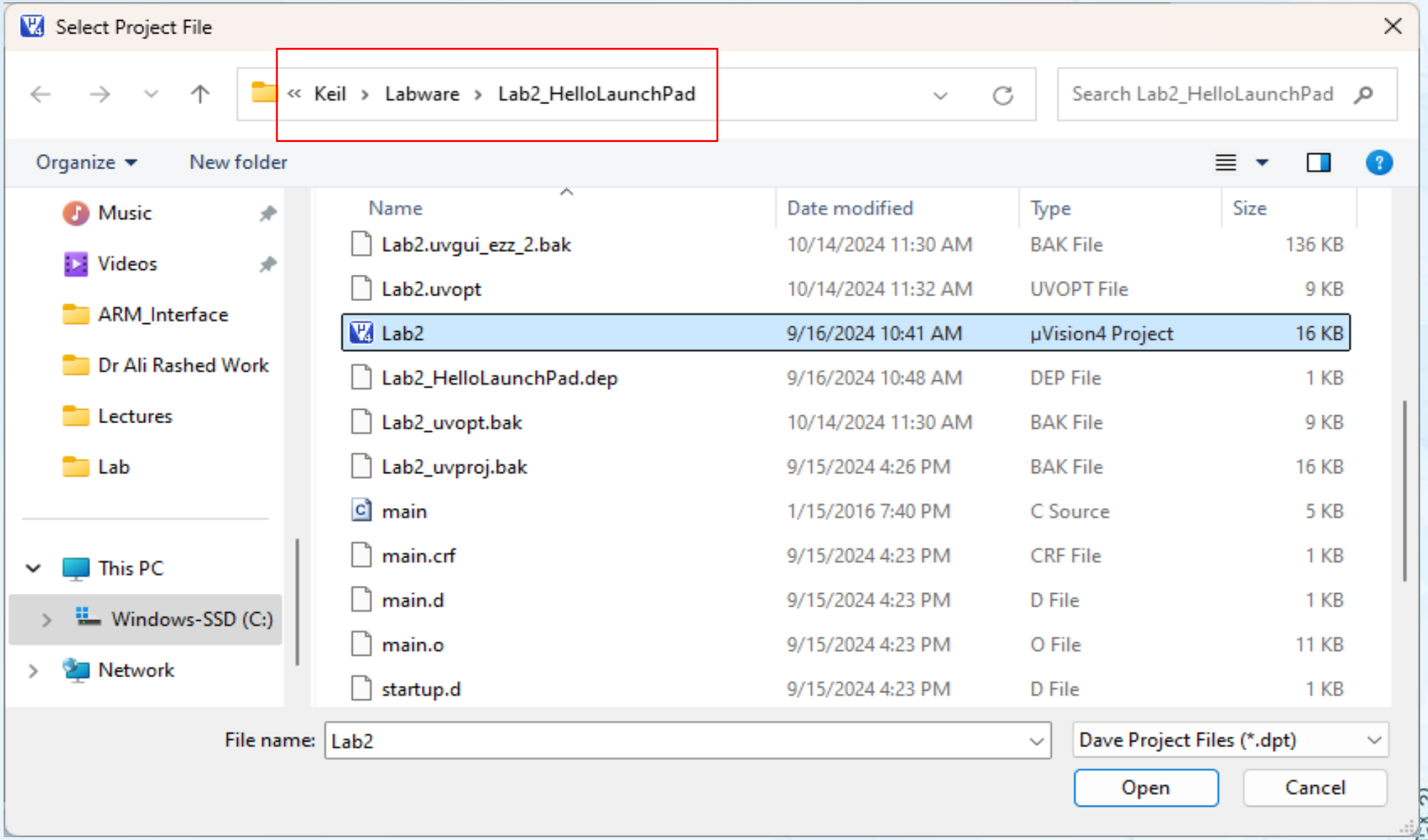


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Second Step: Open Project

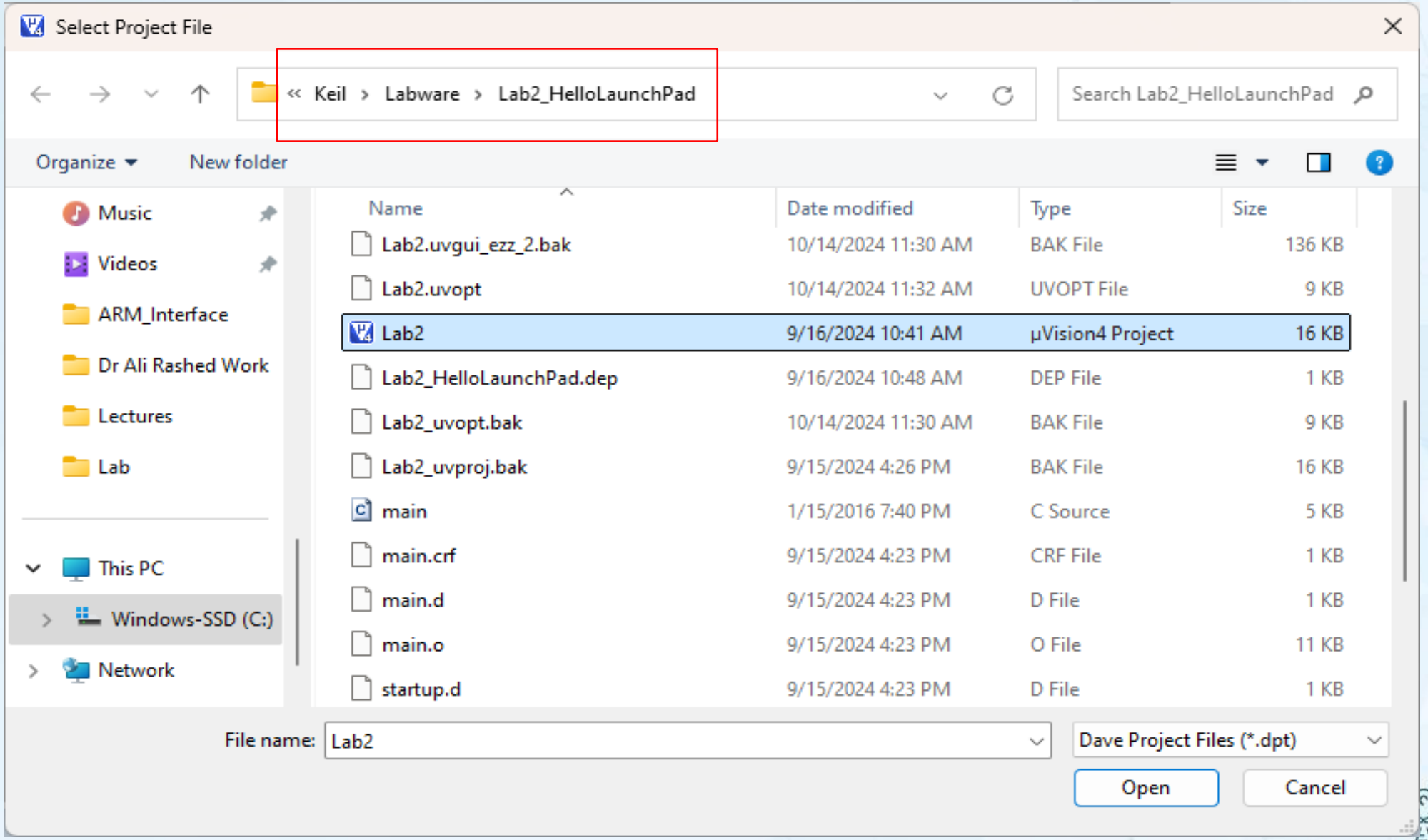


Third Step: Open Project Lab2



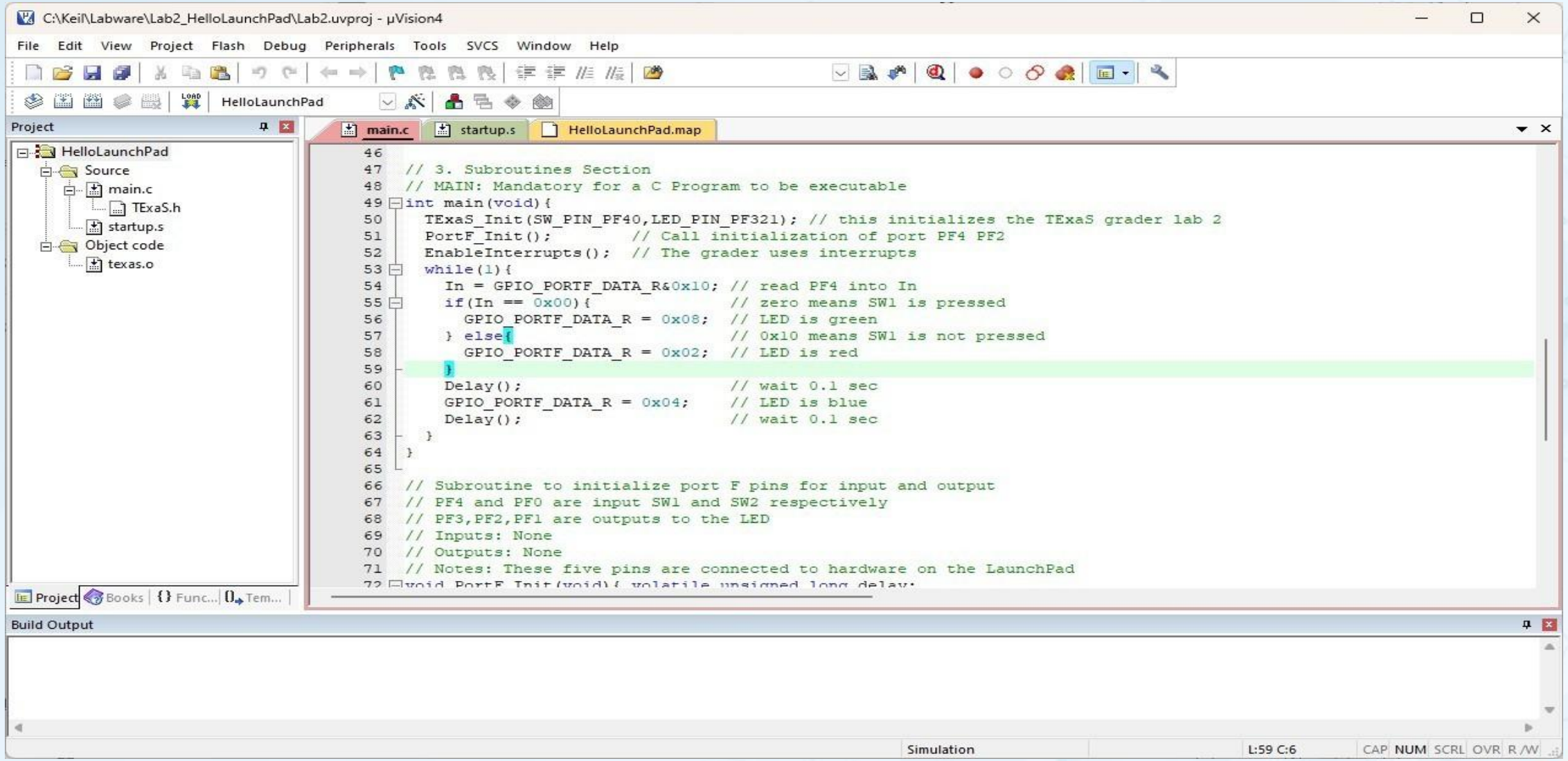
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Third Step: Open Project Lab2

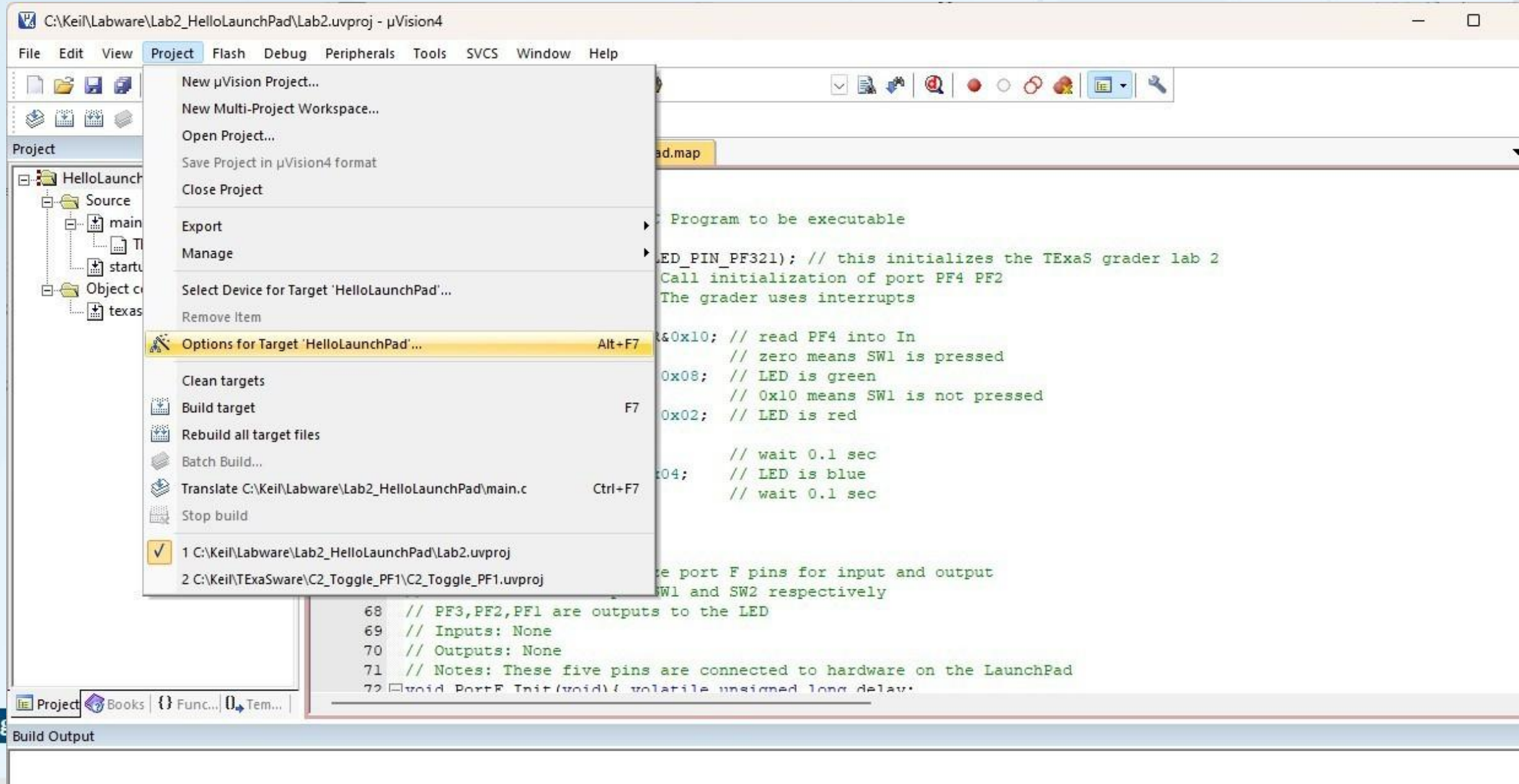


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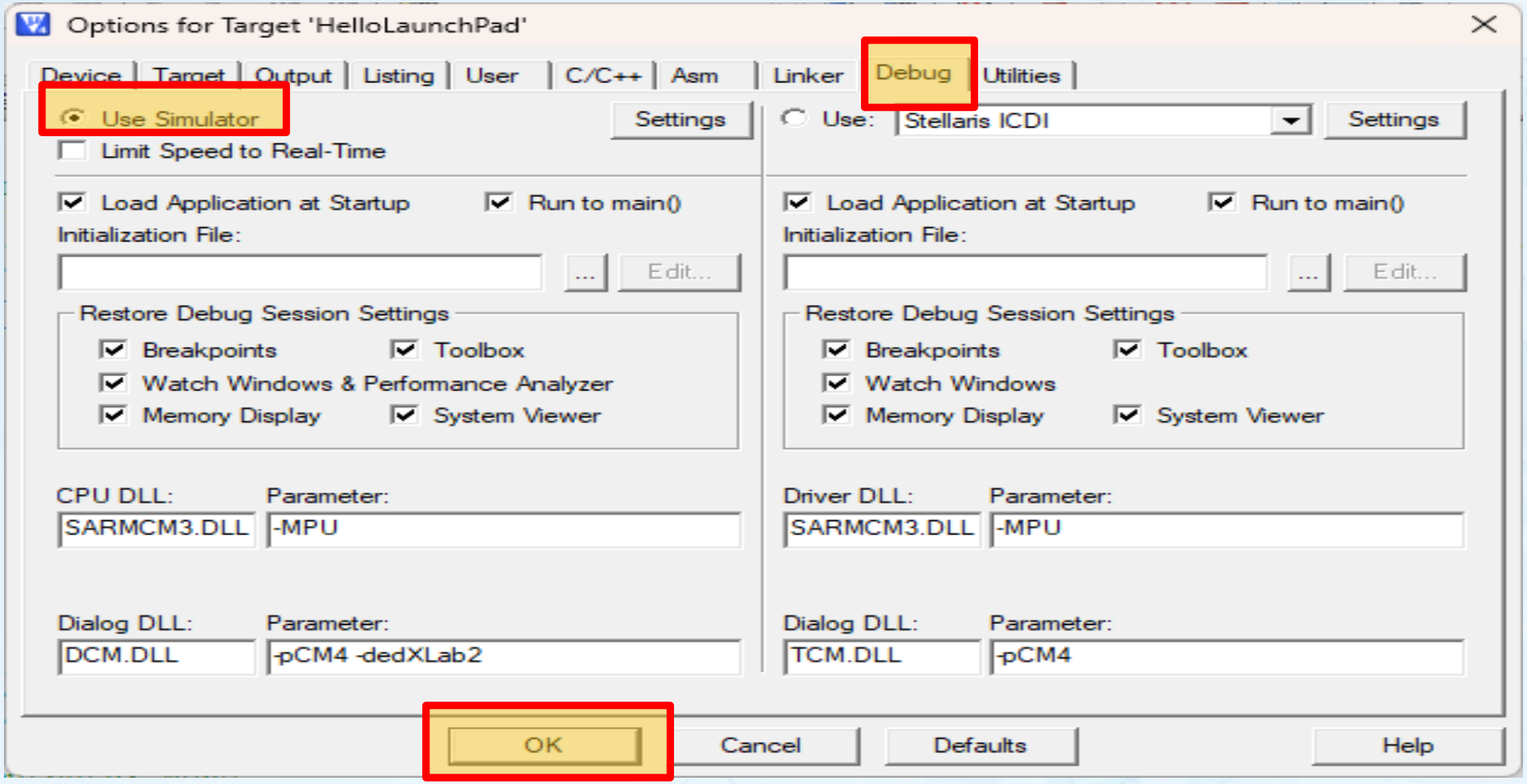
main.c


C:\Keil\Labware\Lab2_HelloLaunchPad\Lab2.uvproj - µVision4
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
HelloLaunchPad
Project
HelloLaunchPad
Source
main.c
TExaS.h
startup.s
Object code
texas.o
main.c
startup.s
HelloLaunchPad.map
46
47 // 3. Subroutines Section
48 // MAIN: Mandatory for a C Program to be executable
49 int main(void){
50 TExaS_Init(SW_PIN_PF40,LED_PIN_PF321); // this initializes the TExaS grader lab 2
51 PortF_Init(); // Call initialization of port PF4 PF2
52 EnableInterrupts(); // The grader uses interrupts
53 while(1){
54 In = GPIO_PORTF_DATA_R&0x10; // read PF4 into In
55 if(In == 0x00){ // zero means SW1 is pressed
56 GPIO_PORTF_DATA_R = 0x08; // LED is green
57 } else{ // 0x10 means SW1 is not pressed
58 GPIO_PORTF_DATA_R = 0x02; // LED is red
59 }
60 Delay(); // wait 0.1 sec
61 GPIO_PORTF_DATA_R = 0x04; // LED is blue
62 Delay(); // wait 0.1 sec
63 }
64 }
65
66 // Subroutine to initialize port F pins for input and output
67 // PF4 and PF0 are input SW1 and SW2 respectively
68 // PF3,PF2,PF1 are outputs to the LED
69 // Inputs: None
70 // Outputs: None
71 // Notes: These five pins are connected to hardware on the LaunchPad
Project Books Func... Tem...
Build Output
Simulation
L:59 C:6
CAP NUM SCRL OVR R/W

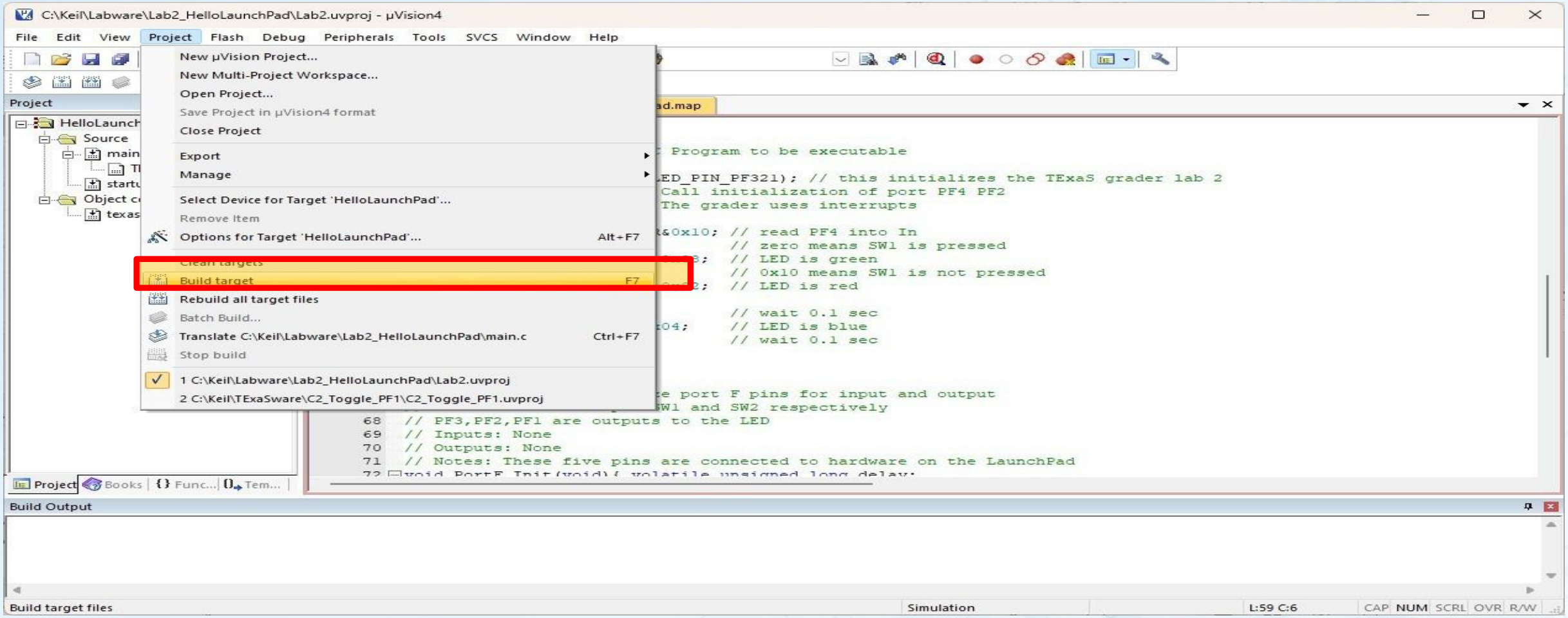
Project->Options for Target



Select Debug -> Use Simulator -> OK



Build The project



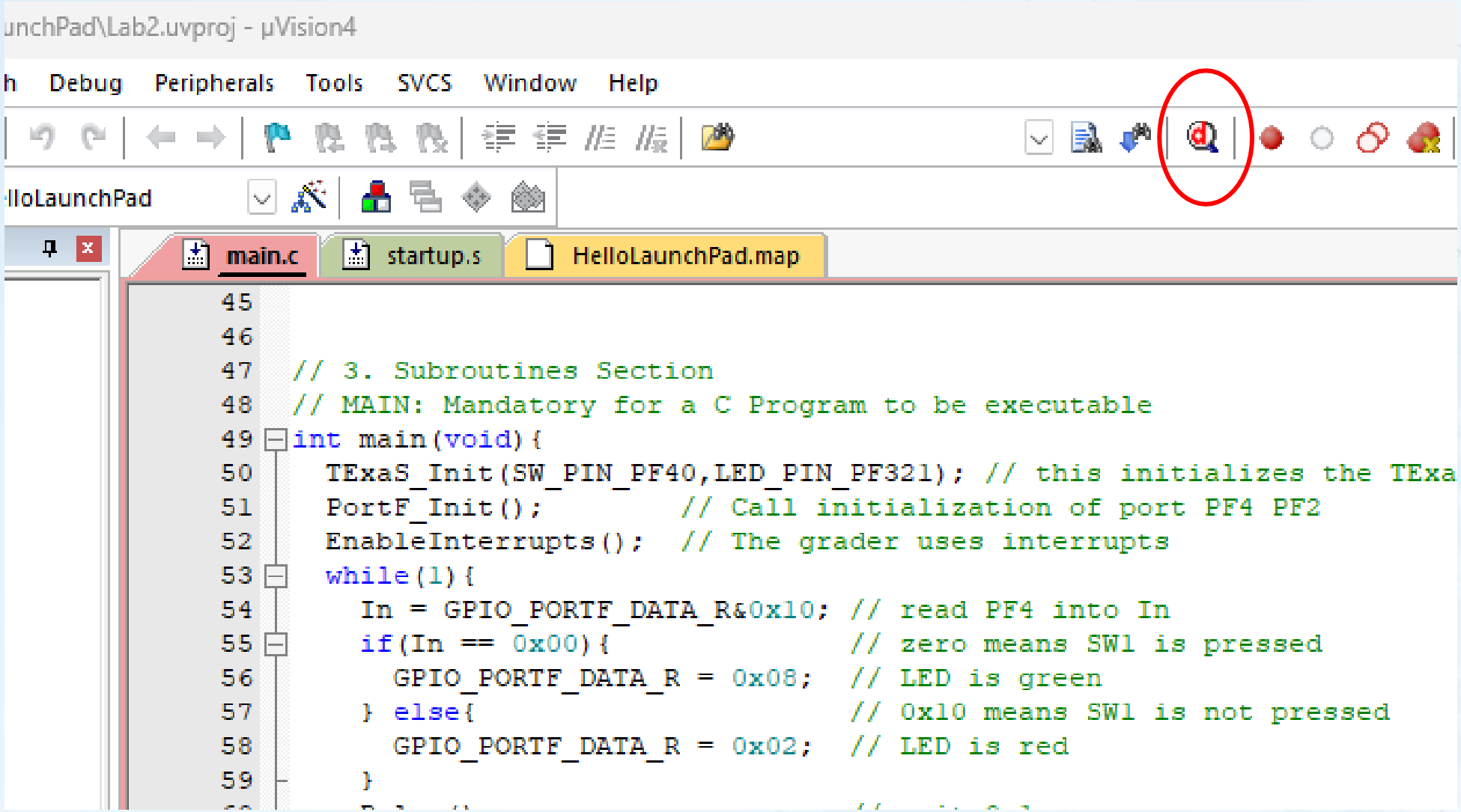
Build The project

Build Output

```
Build target 'HelloLaunchPad'
linking...
Program Size: Code=2732 RO-data=2064 RW-data=52 ZI-data=1756
".\HelloLaunchPad.axf" - 0 Error(s), 0 Warning(s).
```

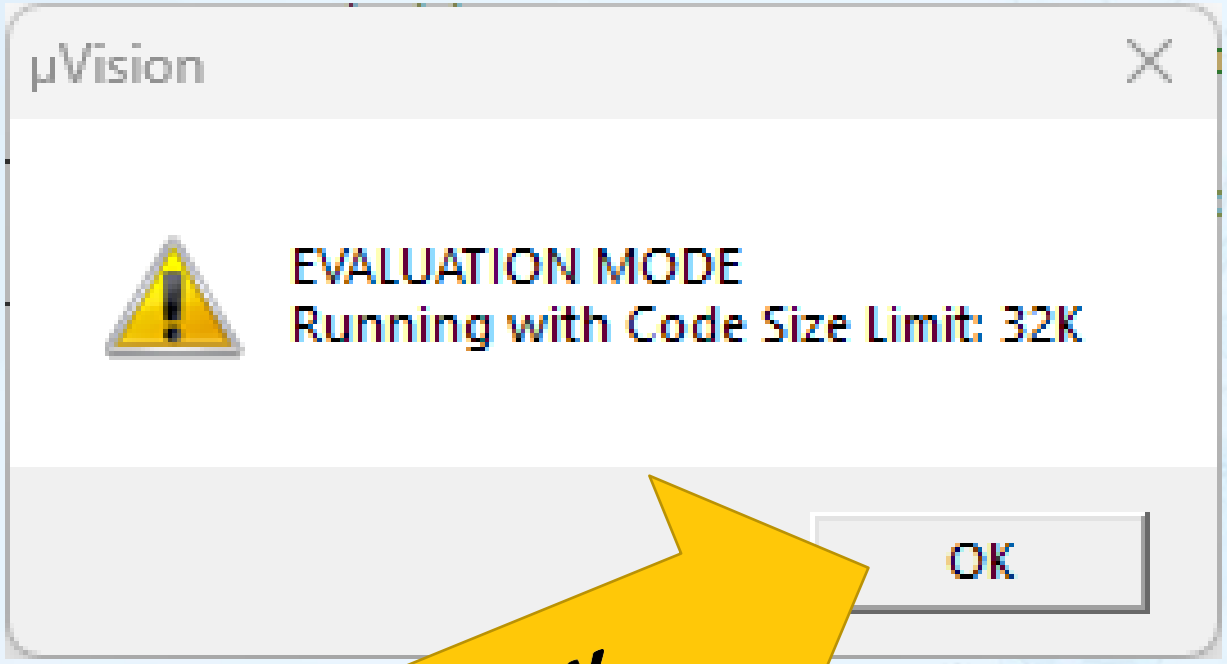

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Start debugging session (CTL+F5)



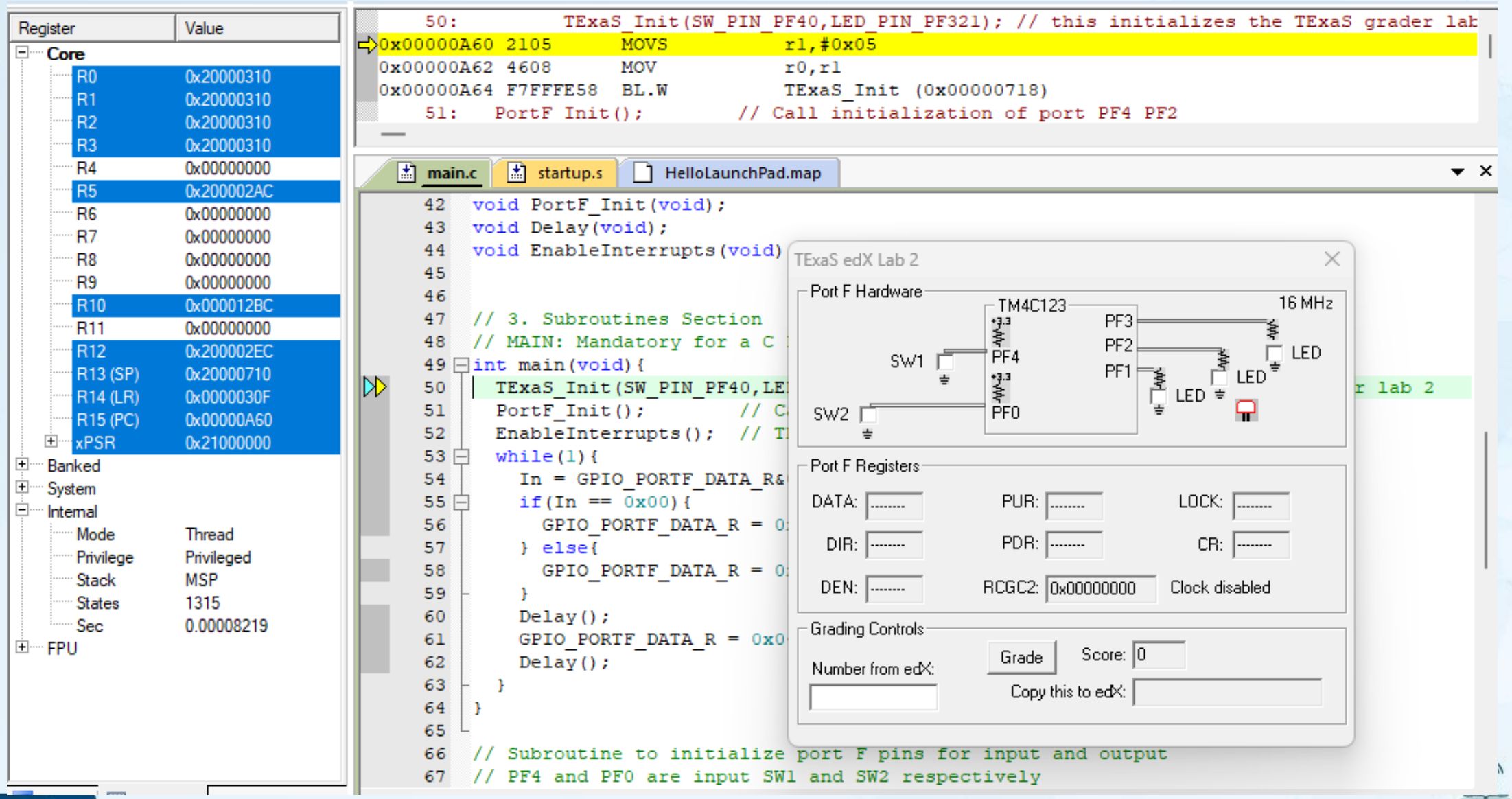
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Start debugging session (CTL+F5)

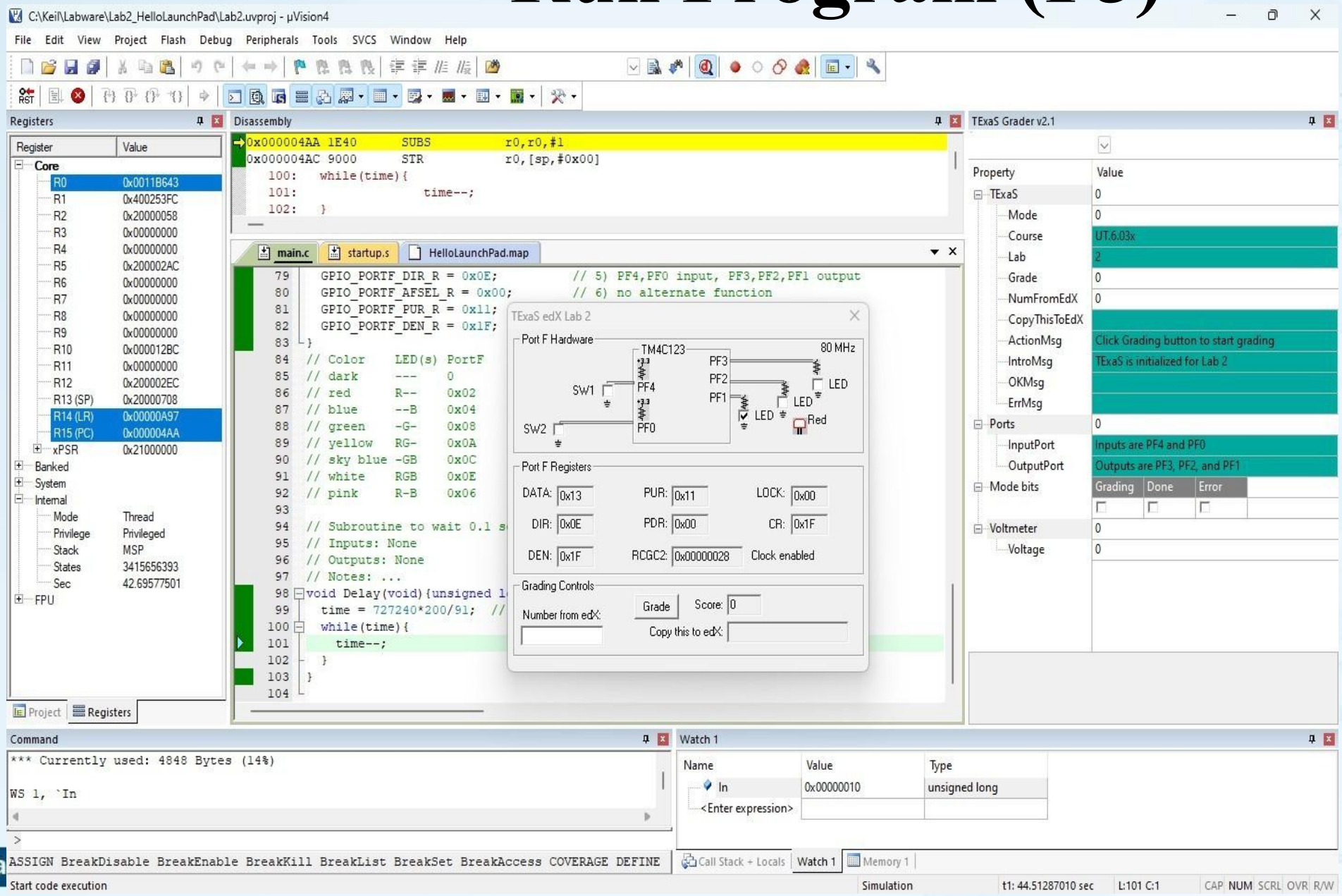


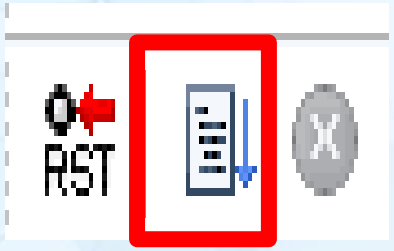
This is a student version so there is a limitation on code size

Start debugging session (CTL+F5)

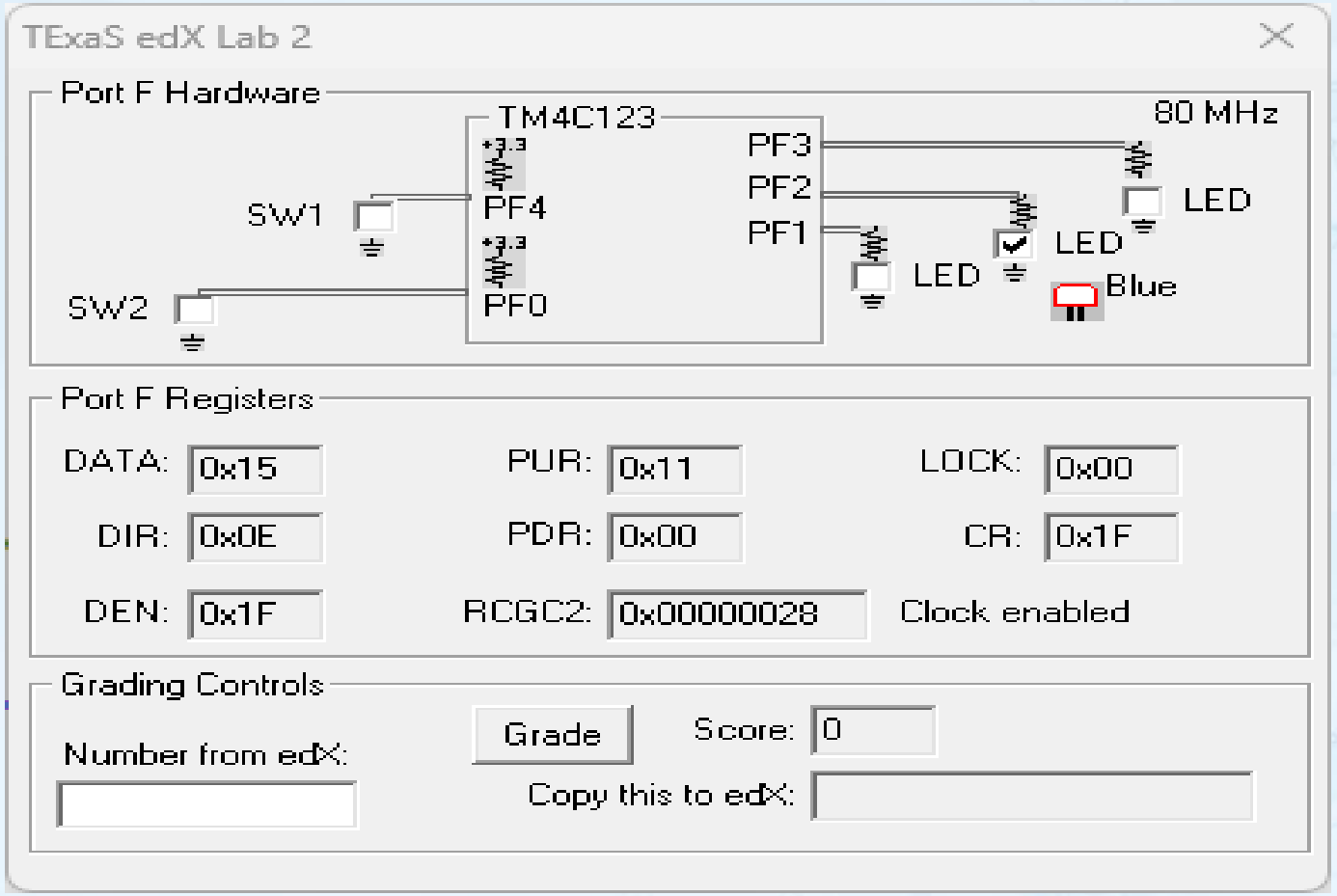


Run Program (F5)





Run Program (F5)

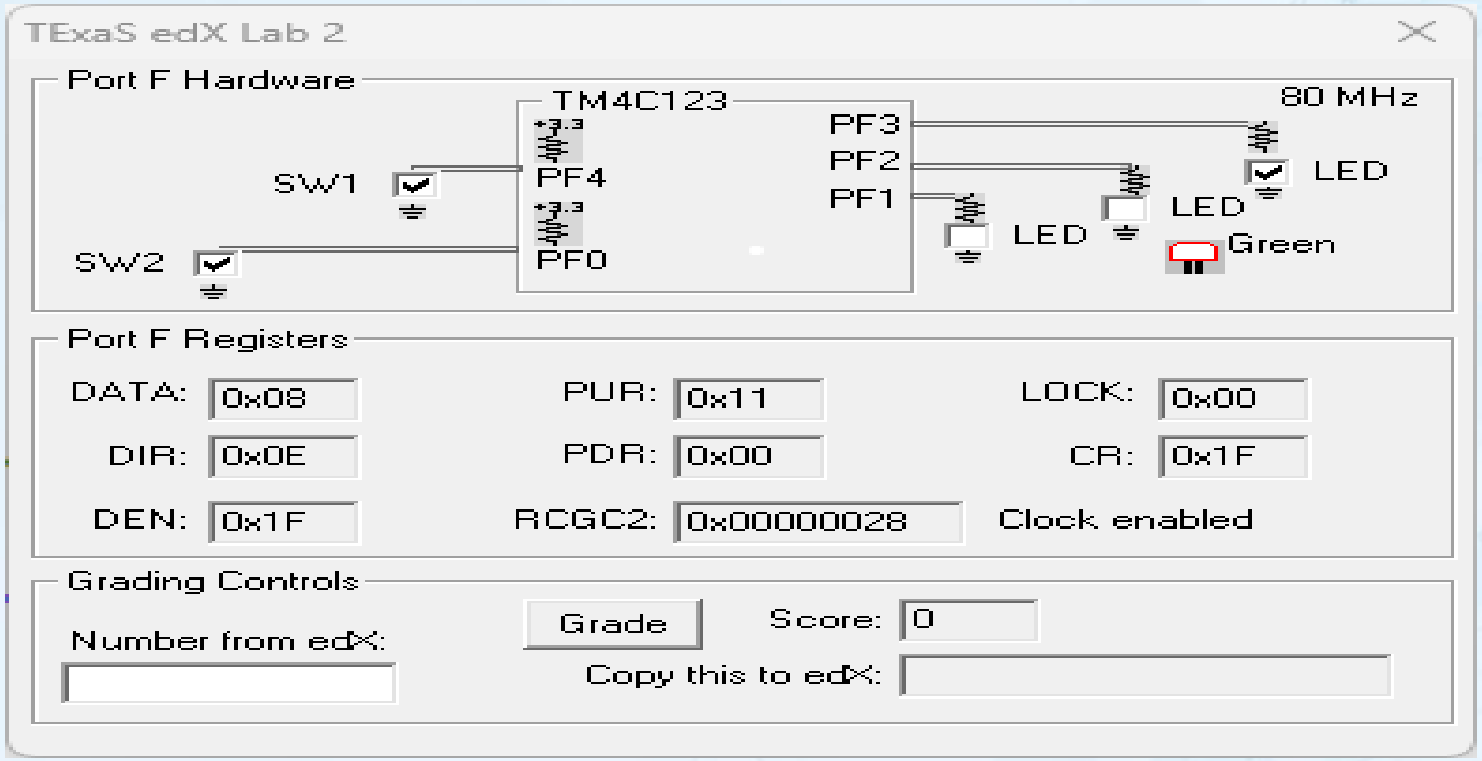


The LED Blue and red are toggle

Run Program (F5)

Press on SW1

The Green Blue and red are toggle



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THANK YOU FOR WATCHING

QUESTIONS?

